The vecmeson generator vs. vpK

A like-for-like comparison for diffractive vector-meson electroproduction

Reaction: $e\,p \to e'\,p'\,V,\quad V \to h^+h^- \quad (V = \rho^0/\phi)$ **Prepared for:** H. Avakian **Status:** stage 5 of 5

Summary. Both generators are driven by the same object—the Diehl unpolarised structure functions $u^{\nu\nu'}_{\mu\mu'} = T\,T^\dagger$—and the same angular intensity $W(\Omega)$. Fed *identical* inputs, with all seven observables rebuilt from the output four-vectors by one common routine, the kinematics and the production angle Φ agree, but the **decay angles reveal a frame-convention difference**: vecmeson-gen uses the standard *s-channel helicity* frame (meson direction in the γ^*p CM); vpK uses the meson direction *in the lab*. Each is clean only in its own frame. The helicity frame is the one the W-kernels are defined in, so it is the correct one for SDME / u extraction. After matching the flux (§7) and reconciling vpK's decay to the CM axis (§9), the two agree across *all seven* observables for the complete natural-parity amplitude set.

1 The two generators

Both simulate $\gamma^*p \to Vp$ followed by $V \to h^+h^-$, weight events by the virtual-photon flux times the helicity-dependent decay intensity, and write CLAS12/GEMC Lund files. They were written independently and share no code.

	vpK (`gagrho`)	**vecmeson-generator**
Language	C++17, no external libraries	Python / NumPy
Physics input	the u-matrix set directly, per element (`generator.h`)	helicity amplitudes $T_{\mu\nu}$; $u = T\,T^\dagger$ built internally
Angular intensity	$W_{UU} = \cos^2\theta\,\mathrm{LL} + \sqrt{2}\sin\theta\cos\theta\,\mathrm{LT} + \sin^2\theta\,\mathrm{TT}$ (`w_kernels.hpp`)	Diehl–Sapeta sum $W = (\sigma_T{+}\varepsilon\sigma_L)A + \sum_k g_k f_k$ (`diehl_w.py`)
Decay-angle frame	meson direction in the **lab**	meson direction in the γ^*p **CM**
Output	Lund, 7 particles + header angles	Lund, 4 particles + header kinematics

2 The shared physics

For unpolarised beam and target and nucleon-helicity non-flip, the amplitude to make a meson of helicity ν from a photon of helicity μ is $T_{\nu\mu}$. The measurable decay distribution is governed by the bilinears (Diehl & Sapeta, *Eur. Phys. J. C* **52** (2007) 933):

$$u^{\nu\nu'}_{\mu\mu'} = \tfrac{1}{2}\sum_\lambda \Big(T_{\nu\mu,\lambda}\,T^*_{\nu'\mu',\lambda} + (\nu,\mu \leftrightarrow -\nu,-\mu)\Big), \qquad \sigma_L \propto u^{00}_{00}, \qquad \sigma_T \propto u^{++}_{++}. \tag{1}$$

The unpolarised decay intensity is a fixed linear combination of these u's and the decay-angle functions f_k:

$$W(\cos\theta,\varphi,\Phi) = (\sigma_T + \varepsilon\,\sigma_L)\,A(\theta) + \sum_k g_k(u,\varepsilon)\,f_k(\theta,\varphi,\Phi). \tag{2}$$

vpK writes this through the W_{UU} kernels; vecmeson-gen writes the equivalent Diehl–Sapeta sum. **Same physics, two encodings.** That is exactly what makes a numerical cross-check meaningful: if the inputs match, the outputs must agree observable by observable.

3 Method

To compare the machinery rather than two models, both generators are given the **same** u-matrix, the **same** windows ($E = 10.6$ GeV, $Q^2 \in [1, 9]$, $x_B \in [0.09, 0.68]$, $|t| \le 5.5$), and the same meson (ρ^0). The study walks three inputs:

Config	Non-zero amplitudes	Expected decay distribution
full model	vpK default u-matrix (all elements)	full angular structure
pure T_{00}	only u_{00}^{00} (longitudinal)	$W \propto \cos^2\theta$, flat in φ, Φ
pure T_{11}	only u_{++}^{++} (transverse)	$W \propto \sin^2\theta$

All seven observables are recomputed from the output four-vectors with a *single* routine applied to both files, so no per-generator header convention can bias the comparison:

$$Q^2 = -(k - k')^2, \qquad x_B = \frac{Q^2}{2\,P\cdot q}, \qquad W = \sqrt{(q + P)^2}, \qquad |t| = -(q - p_V)^2. \tag{3}$$

The two decay angles and the Trento angle Φ are built in the meson rest frame; the *choice of quantization axis* to reach that frame is the subject of stage 2.

4 Stage 1 — pipeline validation

Before any physics claim, the common reader was checked end to end: it parses both Lund formats (7-particle vpK, 4-particle vecmeson-gen), reconstructs the beam and target from the header, and returns all seven observables in physical ranges for both generators. The production angle Φ and Q^2 already overlap at this stage. With the reader trusted, we set the first matched input.

5 Stage 2 — pure longitudinal, and the frame check

Setting only $u_{00}^{00} \neq 0$ is the cleanest possible benchmark: the intensity collapses *analytically* to

$$W(\cos\theta, \varphi, \Phi) = \frac{3}{4\pi}\cos^2\theta \qquad \text{(no } \varphi \text{ or } \Phi \text{ dependence at all).} \tag{4}$$

Every φ- or Φ-modulating term carries a u-element that is zero here, so a correct generator must return an *exact* $\cos^2\theta$ and a *flat* φ. Both generators reproduce their own header angle perfectly—but when the decay angle is rebuilt from the four-vectors, the result depends on which frame you build it in.

The polar angle recovers the analytic shape in the matching frame (density in $|\cos\theta|$ bins, normalised to $1.5\cos^2\theta$):

$\lvert\cos\theta\rvert$ bin	0.1	0.3	0.5	0.7	0.9	convention
$1.5\cos^2\theta$ (exact)	0.015	0.135	0.375	0.735	1.215	—
vecmeson-gen — helicity	0.02	0.14	0.38	0.74	1.21	matches ✓
vpK — helicity	0.23	0.29	0.44	0.64	0.90	pedestal
vpK — lab	0.02	0.14	0.38	0.75	1.22	matches ✓

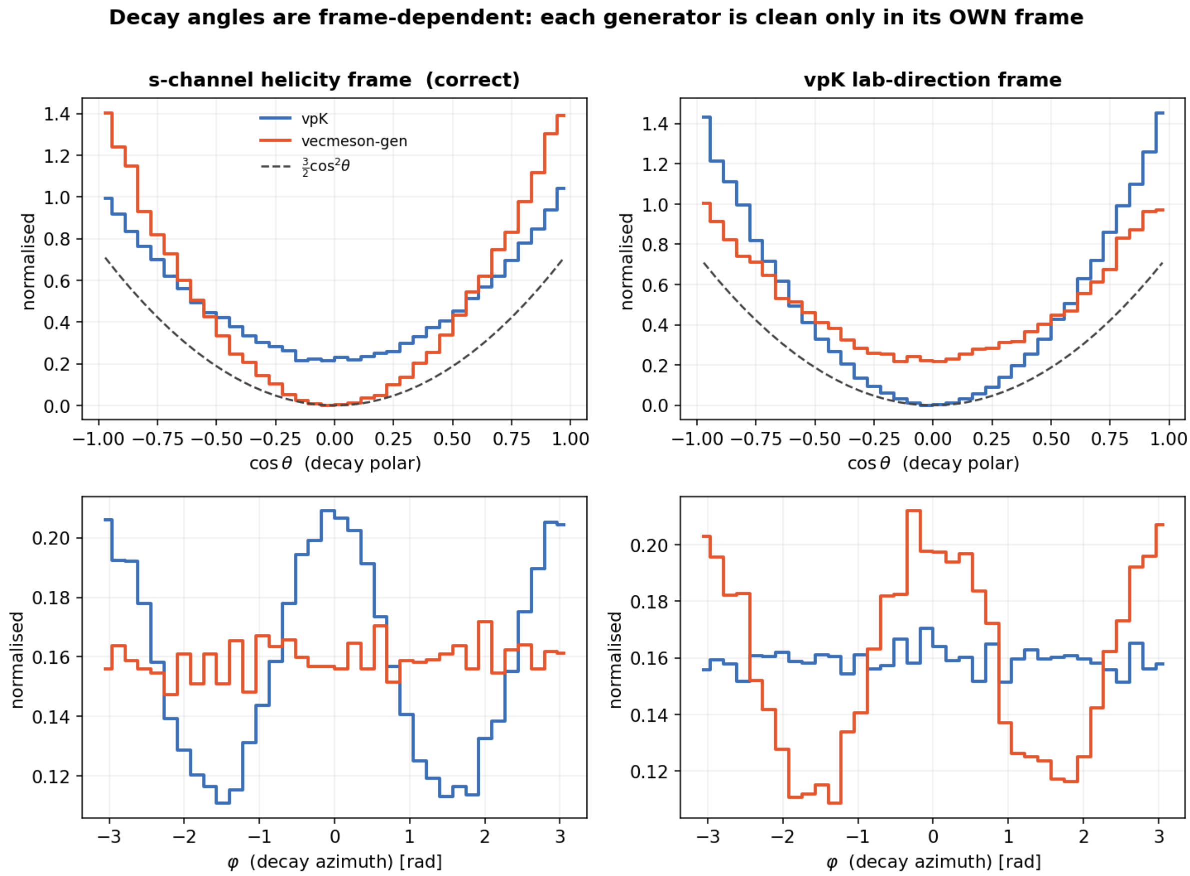

Figure 1. Decay angles for the pure-T_{00} sample, rebuilt from the four-vectors. **Left:** s-channel helicity frame—vecmeson-gen is exact $\cos^2\theta$ and flat φ; vpK carries a pedestal. **Right:** vpK's lab frame—the roles swap. Each generator is clean *only in its own frame*.

The azimuth gives an even sharper single-number test. For pure longitudinal every φ moment must vanish; the frame mismatch shows up as a spurious $\langle\cos 2\varphi\rangle$:

	helicity frame	lab frame
vecmeson-gen $\langle\cos 2\varphi\rangle$	$+0.006 \pm 0.010$ (flat ✓)	$+0.197 \pm 0.010$
vpK $\langle\cos 2\varphi\rangle$	$+0.151 \pm 0.003$	$+0.001 \pm 0.003$ (flat ✓)

A flat azimuth maps into a $\cos 2\varphi$ pattern under the wrong quantization axis because the lab $\leftrightarrow$ CM meson-direction rotation is anisotropic about the production plane—a purely geometric consequence of the frame choice, no new physics.

6 Which frame is correct

The W-kernels that *both* generators use are derived in the **s-channel helicity frame** (Schilling & Wolf 1973; Diehl–Sapeta 2007): the quantization axis is the meson momentum in the $\gamma^* p$ centre-of-mass.

s-channel helicity — correct. $\hat{z} = \hat{p}_V$ in the $\gamma^* p$ CM. Our generator; the frame the SDMEs / u's are defined in.

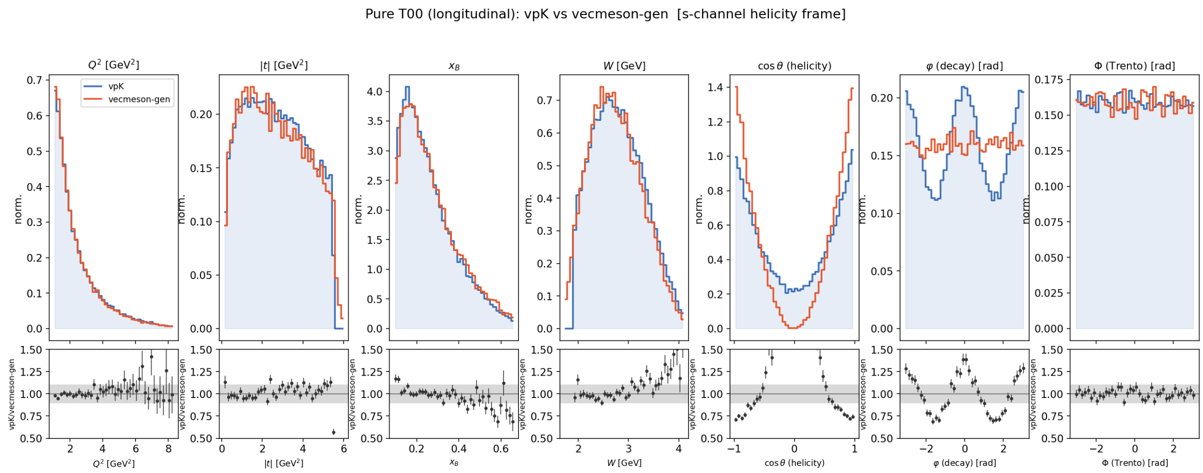

Figure 2. All seven observables, **s-channel helicity frame** (after the flux match of §7). The kinematics $(Q^2, |t|, x_B, W)$ and Φ agree (ratios $\sim$1); only the decay angles $\cos\theta, \varphi$ carry vpK's frame offset (§6).

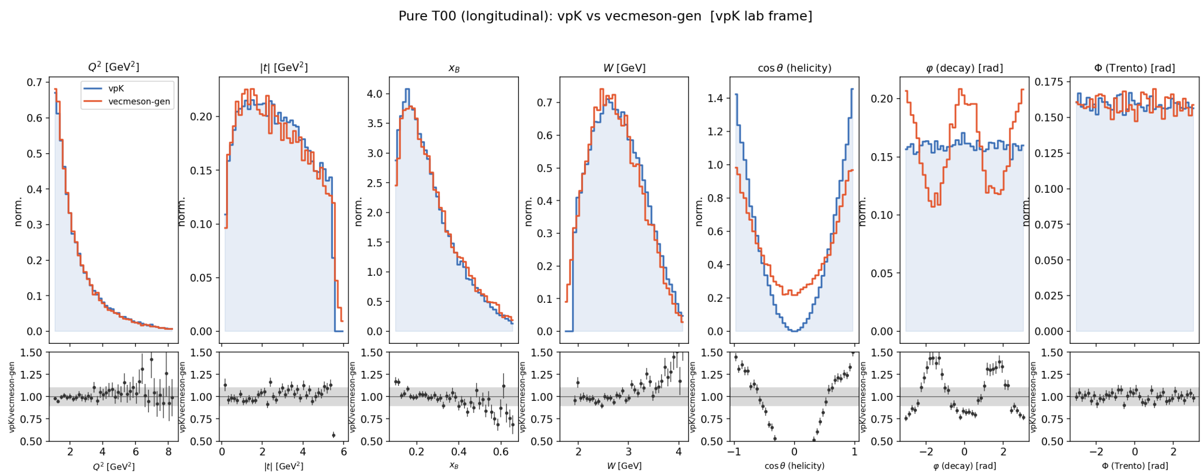

Figure 3. The same, **vpK lab frame**. The decay-angle agreement now flips to vpK, confirming the difference is purely the frame convention.

vpK lab. $\hat{z} = \hat{p}_V$ in the lab (`decay2body.h`: `uz = parent.vect()`). Self-consistent, but a different axis.

Because vpK feeds its lab-frame angle into a kernel derived for the CM axis, a pure-longitudinal sample is $\cos^2\theta$ in vpK's own frame but acquires a $\sin^2\theta$ pedestal and a $\cos 2\varphi$ term when expressed in the helicity frame. The two axes coincide at forward production and separate as $|t|$ grows, so the effect is largest exactly where the transverse/longitudinal separation is done.

Suggested reconciliation. Align vpK's decay basis to the meson direction in the $\gamma^* p$ CM (boost the daughter build into the CM before setting `uz`), so the generated angles live in the same frame its own kernels assume. After that the two generators can be compared on physics alone.

7 Kinematic dependencies — diagnosis and fix

With the u-matrix held constant (pure T_{00}), every non-angular difference is *kinematics*. Rebuilding all seven observables and the drivers $(y, \varepsilon, x_B, m_V)$ isolates the causes; all but the meson lineshape are now resolved:

#	Observable	Status	Root cause / fix
1	Q^2	agrees	both weights $\propto 1/Q^2$
2	Φ (Trento)	agrees (flat)	no interference for pure T_{00}
3	$\cos\theta, \varphi$	**resolved**	frame offset (helicity vs lab axis); vpK patched to the CM axis — §9
4	$y,\ \varepsilon$	**fixed**	flux: `WEIGHT=vpk` reproduces vpK's $d\sigma_{\text{3fold}}$
5	x_B	**fixed** (via #4)	was flux-driven
6	W	**fixed** (via #4)	downstream of x_B
7	$\lvert t\rvert$	**fixed**	`tprime<TMAX`; the $2q^*p^*$ Jacobian cancels
8	m_V	open (minor)	*both* rel. BW, pole M_0, Γ match — only diff is the Blatt–Weisskopf barrier (§9)

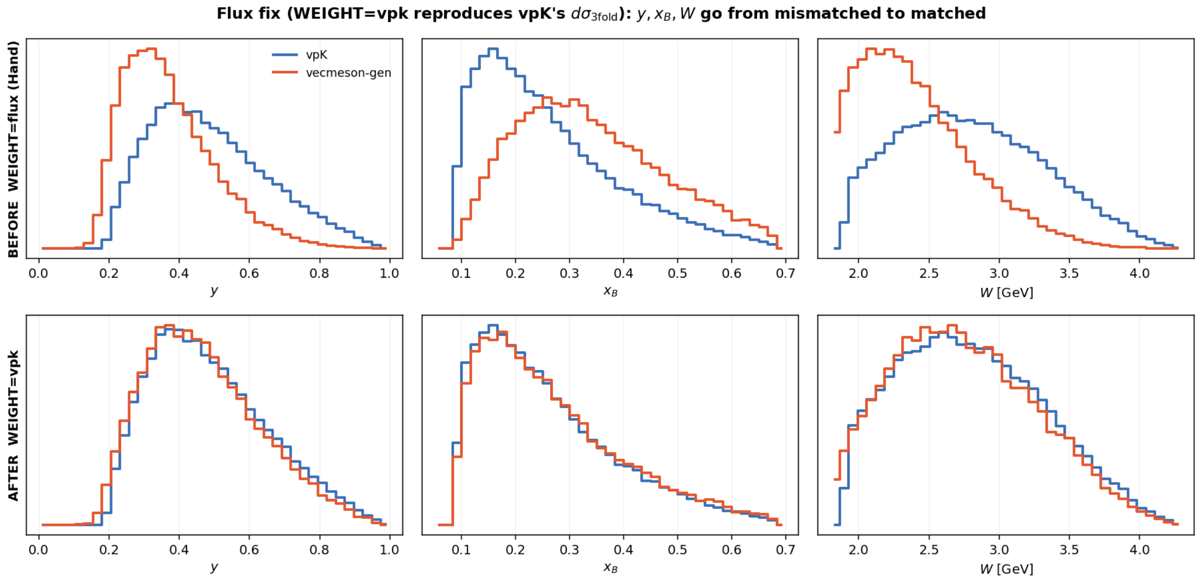

Figure 4. The flux fix. **Top:** with the Hand flux, vecmeson-gen y, x_B, W are shifted from vpK. **Bottom:** the new `WEIGHT=vpk` mode reproduces vpK's $d\sigma_{\text{3fold}}$ and the three overlap.

The dominant driver was the **flux**: vecmeson-gen used the Hand transverse flux $\Gamma_T \propto (1-y)K/(Q^2(1-\varepsilon))$; vpK uses $d\sigma_{\text{3fold}} \propto y^2/(1-\varepsilon)\cdot(1-x_B)/x_B\cdot(1/Q^2)(1+\varepsilon)$. The y-dependence is essentially opposite, so vpK sat at higher y, lower ε, lower x_B, higher W. Adding a `WEIGHT=vpk` mode brings y, ε, x_B, W and $|t|$ into agreement (ratios $\sim$1). The $|t|$ items resolved with it: (a) the hardcoded `tprime<4` is fixed to `tprime<TMAX`; (b) vpK's $2q^*p^*$ factor only compensates its uniform-$\cos\theta^*$ sampling and cancels—vecmeson-gen samples t' flat, so none is needed. Two residuals remain: the meson lineshape (#8, minor) and vecmeson-gen' $|t|$ tail past vpK's $|t| \leq 5.5$ cut. For the upcoming u-matrix port, mind the t-sign (vpK Mandelstam $t < 0$, vecmeson-gen $|t|$). All logged with code references in `ISSUES_vpk_comparison.md`.

8 Amplitude sweep — the full natural-parity set

With the flux matched (§7) and each amplitude turned on alone (constant; in vecmeson-gen a single T, in vpK the corresponding $u = TT^\dagger$ elements, generated from the same `build_T` so the two are identical by construction), the full set was run one at a time. The decay polar angle follows the **meson** helicity, and **vpK and vecmeson-gen agree for every amplitude** (shown in native frames; the frame offset of §6 is a separate effect):

amplitude	meson hel.	$\cos\theta$	$\langle\cos 2\varphi\rangle$	vpK vs vecmeson-gen
T_{00}	0	$\cos^2\theta$	0	agree
T_{01}	0	$\cos^2\theta$	0	agree
T_{11}	±1	$\sin^2\theta$	0	agree
T_{10}	±1	$\sin^2\theta$	$+0.50$	agree
T_{1-1}	±1	$\sin^2\theta$	0	agree

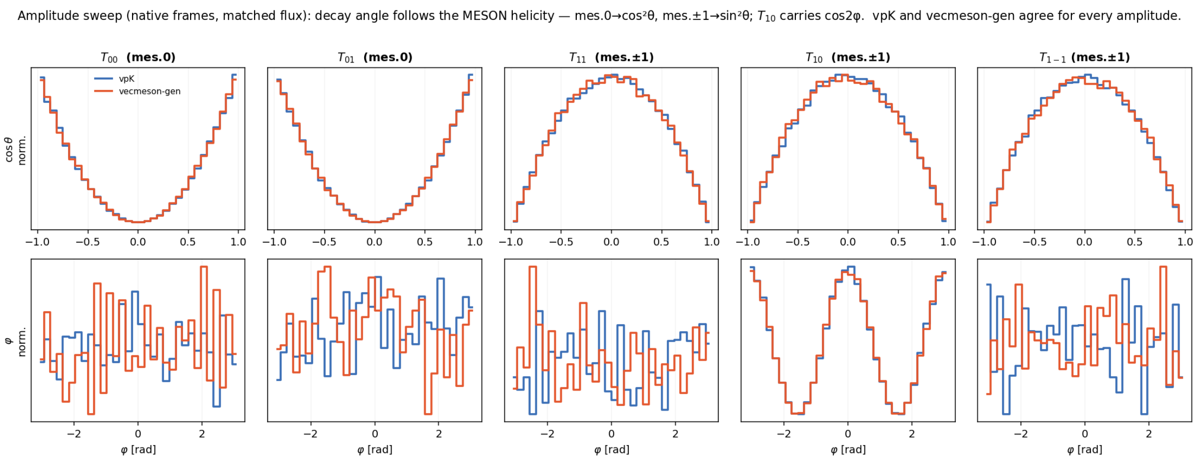

Figure 5. The five natural-parity amplitudes, one at a time (vpK vs vecmeson-gen, native frames, matched flux). **Top:** $\cos\theta$ — meson-longitudinal amplitudes (T_{00}, T_{01}) give $\cos^2\theta$, meson-transverse (T_{11}, T_{10}, T_{1-1}) give $\sin^2\theta$. **Bottom:** φ — flat except T_{10}, which carries a clear $\cos 2\varphi$. Every vpK curve overlaps its vecmeson-gen counterpart.

9 Frame reconciliation — vpK in the helicity frame

Item #3 — the one genuine physics discrepancy — is a convention, not a bug: vpK builds its decay about the meson direction in the *lab* (`decay2body.h`), whereas the Diehl kernels are defined about the $\gamma^* p$-CM (helicity) direction. Patching vpK's `produce__3` to build the decay about the meson momentum *in the CM* (reusing `decay()` unchanged, fed the CM-boosted meson, daughters boosted back to the lab) removes the offset entirely:

Repeating the full sweep with the patched vpK, **both generators now agree on the decay angles too** — $\cos\theta$ and φ, for every amplitude, in one common (correct) frame:

With the flux matched (§7) and the decay frame reconciled, vpK and vecmeson-gen agree across *all seven* observables for the complete natural-parity set. The change is isolated to `produce__3` (in a working copy, not H.A.'s original) and is offered for discussion — the physics case being that the W-kernels are derived in the helicity frame.

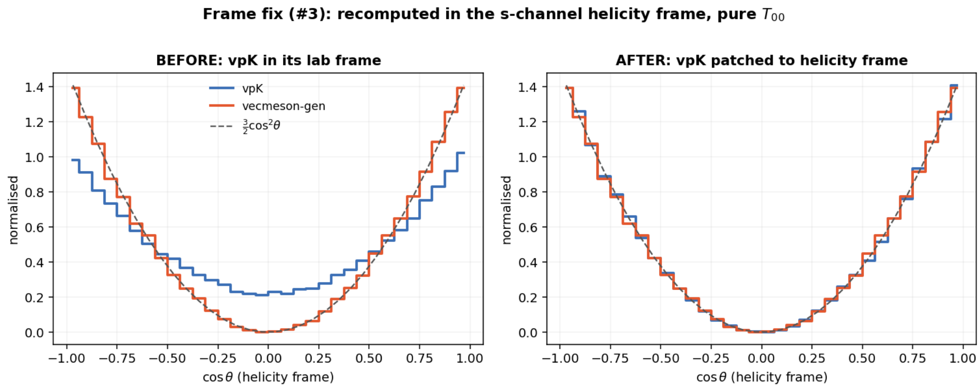

Figure 6. Pure T_{00}, $\cos\theta$ recomputed in the helicity frame. **Left:** vpK in its lab frame carries the $\sin^2\theta$ pedestal. **Right:** with vpK's decay basis moved to the CM meson direction, vpK and vecmeson-gen both collapse onto the exact $\frac{3}{2}\cos^2\theta$.

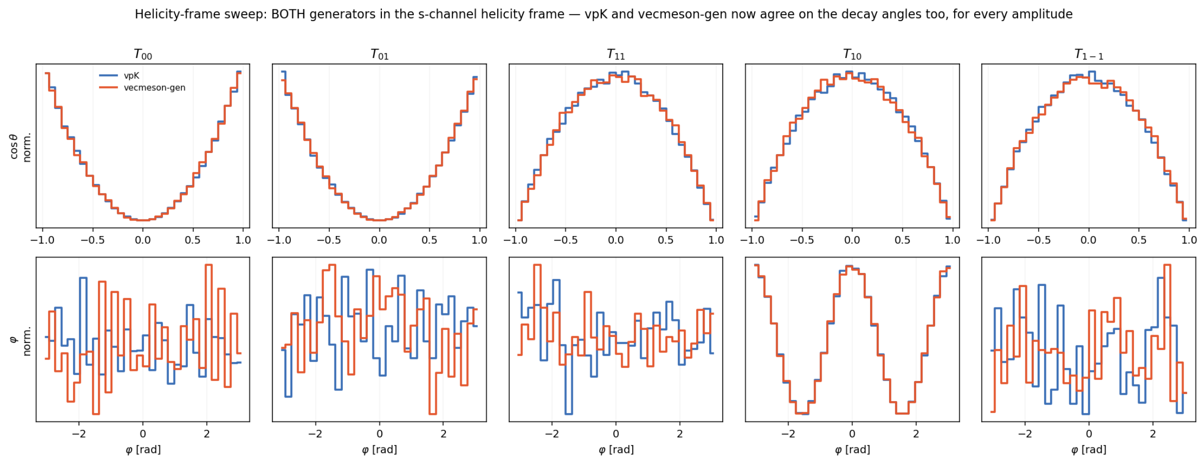

Figure 7. The five amplitudes with *both* generators in the s-channel helicity frame. Every vpK curve overlaps its vecmeson-gen counterpart in $\cos\theta$ (top) and φ (bottom): the full angular structure now matches, not just the kinematics.

10 Meson lineshape m_V — for discussion

This is the one open kinematic item (#8). **Both generators use a relativistic Breit–Wigner** with a Jackson P-wave running width $\Gamma(m) = \Gamma_0\,(p/p_0)^3\,(M_0/m)$, and the pole parameters are nearly identical:

	vecmeson-gen (`vecmeson-gen`)	vpK (`gagrho.cpp`)
M_0	0.77526 GeV (current PDG)	0.77 GeV
Γ_0	0.1491 GeV	0.1502 GeV
running width	Jackson × **Blatt–Weisskopf** $B(p)$	bare Jackson
$B(p)$	$\dfrac{1+(p_0R)^2}{1+(pR)^2}$, $R = 5\ \text{GeV}^{-1}$	— (absent)
mass window	$[\,2m_\pi,\ M_0 + 8\Gamma_0\,]$	$[\,2m_\pi,\ \min(W - M_p,\ M_0 + 5\Gamma_0)\,]$

(The README's $M_0 = 0.7683$ is stale; the code sets 0.77.) So the pole mass and width agree to a few MeV — the **only structural difference is the Blatt–Weisskopf barrier**, present in vecmeson-gen and absent in vpK.

For discussion with H.A. The Blatt–Weisskopf barrier accounts for the finite range of the

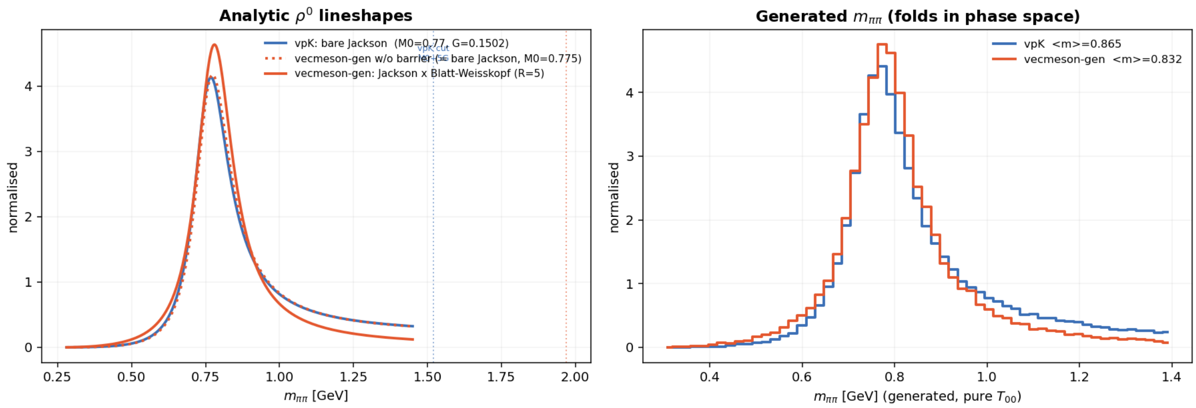

Figure 8. Left: analytic ρ^0 lineshapes. vpK's bare Jackson and vecmeson-gen *without* the barrier coincide (the pole masses match); the Blatt–Weisskopf barrier (vecmeson-gen, solid) sharpens the peak and suppresses the high-mass tail above ~ 0.9 GeV. **Right:** the generated $m_{\pi\pi}$ (folding in phase space): vpK's tail is slightly longer, $\langle m \rangle = 0.865$ vs 0.832 GeV.

$\rho \to \pi\pi$ decay and is the more complete choice; vpK's bare Jackson is the common simpler one — neither is wrong. Points to settle for a joint sample: (i) adopt one running-width form in both (add the barrier to vpK, or a `BW_BARRIER=0` knob in vecmeson-gen); (ii) align the upper mass window ($M_0 + 5\Gamma$ vs $M_0 + 8\Gamma$); (iii) the physics impact is small — once the flux is matched, the ~ 0.03 GeV shift in $\langle m \rangle$ barely moves W or t.

11 Next

All three matched conditions — flux (§7), amplitude-by-amplitude angular structure (§8), and the decay frame (§9) — now agree. Remaining, optional:

- **Full interfering model.** All amplitudes on together (vpK's default u-matrix), exercising the L-T/T-T interference terms and the Φ-dependence, with the t-sign flip.
- **Residual kinematics.** Settle the m_V lineshape (§10) and apply vpK's $|t| \leq 5.5$ cut.
- **Statistics.** Scale each config to $\gtrsim$ 1M events for the final canvases.

All seven observables rebuilt from Lund four-vectors with one routine (`compare.py`). Sample: pure-T_{00}, ρ^0, $E = 10.6$ GeV, matched windows; vpK 60k, vecmeson-gen 55k events. Location: `/w/ceph24/.../Softwares/Test_vecMeson_gen` on ifarm.